

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-CT-7801-03 Sheet 1 of 1
OPL Title	Fan Blade Inspection & Balance (FRP)	Discipline	Mechanical	
Equipment	CT-7801 - COOLING TOWER CELL FAN	Area / Unit	7800 - COOLING TOWER SYSTEM	
Related Interlock	SEQ-7801 (SIL 1)	P&ID; Ref	TJC-LLD-PID-7801	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **fan blade inspection & balance (frp)** for CT-7801 (COOLING TOWER CELL FAN) in the COOLING TOWER SYSTEM. A cracked or eroded FRP blade unbalances the fan and causes the VSHH-7802 vibration trip. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **NON CRITICAL** handling **Circulating water / air** at up to atm / 50 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Lock out the fan and inspect all 6 FRP blades	Isolation & LOTO verified
2	Check for cracks, leading-edge erosion and tip clearance	Parts match BOM / spec
3	Verify the blade pitch angle is uniform on all blades	Torque / clearance to spec
4	Re-balance the fan if vibration is high	No leak / smooth operation
5	Torque the blade root bolts to specification	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled fan gearbox oil change	Routine - oil sample acceptable	Drained and refilled ISO VG 220, replaced breather
VSHH-7802 vibration alarm on CT-7801 fan	Leading-edge erosion on one FRP blade caused imbalance	Inspected blades, re-balanced fan, vibration normalised
TSHH-7803 gearbox oil temperature rising	Degraded oil and fouled oil cooler	Changed oil, cleaned oil cooler, verified temperature

6. KEY LEARNING POINTS

- A cracked or eroded FRP blade unbalances the fan and causes the VSHH-7802 vibration trip.
- Respect the interlock SEQ-7801: e.g. VSHH-7802 trips at > 9 mm/s - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID; TJC-LLD-PID-7801 before working on CT-7801.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

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